

# EXPERIMENT REPORT

## Comparative Study of LSTM and Transformer Models

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### Aim:-

To design and evaluate two deep learning models, namely LSTM and Transformer, for classification of time-series sensor data and analyze their performance differences.

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### Objective:-

- To implement LSTM for sequential data modeling
  - To implement Transformer using attention mechanism
  - To compare both models on the basis of performance metrics
  - To study the impact of architecture on learning efficiency
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### Introduction:-

Time-series data plays an important role in modern machine learning applications such as activity recognition and sensor analysis. Traditional models like LSTM process data sequentially, while newer architectures like Transformers utilize attention mechanisms to improve performance.

This experiment focuses on comparing both approaches using real-world sensor data.

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### Theory:-

#### ◆ LSTM Model

LSTM is an advanced form of recurrent neural network that maintains information over time using gated mechanisms. It is suitable for sequential problems but suffers from slower computation due to its step-by-step processing.

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## ◆ Transformer Model

Transformer eliminates recurrence and instead uses self-attention to capture relationships in the data. It processes data in parallel, making it more efficient and effective in handling complex dependencies.

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### Procedure:-

1. Multiple datasets representing different classes were loaded
  2. Data preprocessing was performed:
    - Removal of missing values
    - Conversion to numeric format
  3. Relevant features from sensor readings were selected
  4. Data normalization was applied
  5. Sequences were generated using a fixed window size
  6. Dataset was divided into training and testing sets
  7. Two models were trained separately:
    - LSTM network
    - Transformer-based model
  8. Performance was evaluated using accuracy and visualization techniques
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### Results

| Model       | Accuracy (Approx.) |
|-------------|--------------------|
| LSTM        | 62–68%             |
| Transformer | 72–78%             |

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**Performance Analysis:-**

- Transformer showed better learning capability due to attention mechanism
  - Training time for Transformer was relatively faster
  - LSTM required more time due to sequential processing
  - Transformer handled patterns in data more effectively
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**Comparison Summary:-**

| Criteria            | LSTM       | Transformer |
|---------------------|------------|-------------|
| Processing Style    | Sequential | Parallel    |
| Learning Efficiency | Moderate   | High        |
| Execution Speed     | Slower     | Faster      |
| Dependency Handling | Limited    | Strong      |
| Model Complexity    | Lower      | Higher      |

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**Conclusion:-**

From the experiment, it is observed that Transformer provides better performance compared to LSTM for time-series classification tasks. The use of attention mechanism enables the model to capture important relationships in the data efficiently.

Although LSTM is effective for sequential data, Transformer proves to be more powerful and scalable for modern applications.

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### **Applications:-**

- **Smart vehicle monitoring**
- **Wearable device data analysis**
- **Motion detection systems**
- **Industrial sensor analytics**

### **My Comments:-**

- **During this experiment, I learned how different deep learning models behave on the same dataset. Initially, the LSTM model gave decent results but required more time to train due to its sequential nature. When I implemented the Transformer model, I observed that it trained faster and achieved better accuracy.**
- **This experiment helped me understand the importance of attention mechanisms and how modern models can outperform traditional ones. It also improved my practical understanding of handling time-series data and comparing different architectures.**
- **Overall, it was a valuable learning experience that enhanced both my coding and analytical skills.**